GRADE 11 · ALGEBRA AND CALCULUS · QUARTER II

Approximation, modelling and the indefinite integral

A workbook to be written in. Each topic runs Focus (easy), Practice (medium), Challenge (hard) and Beyond (very hard, olympiad). Answers, with solutions for the Beyond questions, are at the back.

Topic 1	Approximate calculations with the derivative	lessons 28–30	45 questions
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6 topics · 270 questions · 45 per topic

Approximate calculations with the derivative

The tangent standing in for the curve — good enough for a square root, and only over a short step.

Lessons 28–30 · Algebra and Calculus 11, §1.9 · Pure Mathematics 1, 8.4 · Pure Mathematics 2, 6 · 45 questions

KEY WORDS

- approximation
- linear approximation
- small increment
- differential
- absolute error
- relative error
- percentage error
- convenient point
- order of accuracy

REMEMBER

1. $\Delta y \approx f'(x) \cdot \Delta x$, so $f(x + \Delta x) \approx f(x) + f'(x) \cdot \Delta x$.
2. Choose a convenient point — one where f and f' are exact, and close to the value wanted.
3. $dy = f'(x)dx$ is the rise along the tangent; Δy is the rise along the curve. They differ by an amount of order $(\Delta x)^2$.
4. Relative error is $\left|\frac{\Delta y}{y}\right|$; percentage error is 100 times it.
5. A p % error in a length gives about $2p$ % in an area and $3p$ % in a volume.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $f(x) = x^2$. Estimate Δy at $x = 3$ with $\Delta x = 0.01$.

2. $f(x) = x^3$. Estimate Δy at $x = 2$ with $\Delta x = 0.1$.

3. $f(x) = \sqrt{x}$. Estimate Δy at $x = 9$ with $\Delta x = 0.1$.

4. Estimate $\sqrt{101}$.

5. Estimate $\sqrt{4.02}$.

6. Estimate $(2.01)^2$.

7 Estimate $(1.002)^3$.

8 $f(x) = \dfrac{1}{x}$. Estimate Δy at $x = 2$ with $\Delta x = 0.1$.

9 Estimate $\dfrac{1}{1.01}$.

10 Write down the small-increment formula.

Practice

MEDIUM · 10 questions

The same skill with more to simplify first.

1 Estimate $\sqrt{26}$.

2 Estimate $(8.1)^{1/3}$.

3 Estimate $(3.02)^3$.

4 Estimate $\dfrac{1}{\sqrt{4.04}}$.

5 A square has side 10 cm measured to within 0.1 cm. Estimate the error in its area.

6 For that square, find the percentage error in the area.

7 A circle has radius 5 cm with an error of 0.05 cm. Estimate the error in its area.

8 A cube has edge 6 cm with an error of 0.02 cm. Estimate the error in its volume.

9 Estimate $\sin 0.02$, with the angle in radians.

10 Estimate $\tan 0.01$, with the angle in radians.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Estimate $\sqrt{102}$ to three decimal places.

2 Estimate $(1001)^{1/3}$ to four decimal places.

3 Estimate $\sin 31°$ to three decimal places.

4 Estimate $\ln 1.05$.

5 Estimate $e^{0.02}$.

6 A sphere has radius 10 cm with an error of 0.1 cm. Find the percentage error in its volume.

7 The radius of a circle increases by 2 %. Find the percentage increase in its area.

8 Estimate $(2.01)^4$.

9 Show that $(1 + x)^n \approx 1 + nx$ for small x.

10 Estimate $\sqrt{0.98}$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Show that the error in the linear approximation of f at a is of order $(\Delta x)^2$.

2 Estimate $\sqrt{50}$, and say how far the estimate is from the true value.

3 Use the tangent at $x = 0$ to show $\sqrt{1 + x} \approx 1 + \frac{x}{2}$, and test it on $\sqrt{1.1}$.

4 Compare the linear estimate of $(1.1)^{10}$ with the true value, and say what the comparison shows.

5 The period of a pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. If L increases by 1 %, find the percentage change in T.

6 A cone has fixed height. If its base radius increases by 3 %, find the percentage increase in its volume.

7 Show that for $f(x) = \frac{1}{x}$ with $x > 0$ the linear approximation always underestimates.

8 Use the iteration $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ with $f(x) = x^2 - 2$ and $x_0 = 1.5$ to find $\sqrt{2}$ to four decimal places.

9 Use the same iteration to solve $x^3 - x - 1 = 0$, starting from $x_0 = 1.5$, to four decimal places.

10 Show that the relative error in x^n is n times the relative error in x.

11 Estimate $\cos 59°$ to four decimal places.

12 The series for $\sqrt{1 + x}$ begins $1 + \frac{x}{2} - \frac{x^2}{8}$. Use the extra term to improve the estimate of $\sqrt{1.02}$.

13 A quantity x is measured with a 1 % error. Find the percentage error in $\frac{1}{x^2}$.

14 Show that the linear estimate of $f(a + h)$ is exact when f is linear.

15 Find how large h may be so that the linear estimate of $(1 + h)^3$ is within 0.01 of the true value.

Modelling with the derivative

A rate in the world, written as a derivative — motion, growth, cost — then read back as an answer with units.

Lessons 31–34 · Algebra and Calculus 11, §1.10 · Pure Mathematics 1, 8.4 · 45 questions

KEY WORDS

- mathematical model
- displacement
- velocity
- acceleration
- connected rates
- marginal cost
- marginal revenue
- units of a rate
- instant of rest

REMEMBER

1. $v = \frac{ds}{dt}$ and $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$. At rest means $v = 0$, not $s = 0$.
2. Connected rates run through the chain rule: $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$. Substitute the numbers last.
3. Displacement is $s(t_2) - s(t_1)$; distance travelled splits the journey at every instant of rest.
4. Marginal cost is $C'(x)$ and marginal revenue $R'(x)$; profit is greatest where they are equal.
5. Every answer carries units, and a negative rate means the quantity is falling.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $s = t^2 + 3t$. Find v.

2. $s = t^2 + 3t$. Find v at $t = 2$.

3. $s = t^3$. Find the acceleration.

4. $v = 3t^2 - 12t$. Find every instant at which the particle is at rest.

5. $s = 5t^2$. Find v at $t = 1$.

6 s is in metres and t in seconds. State the units of $\frac{ds}{dt}$.

7 $C(x) = 100 + 5x$. Find the marginal cost.

8 $R(x) = 20x - x^2$. Find the marginal revenue.

9 $s = 10t - t^2$. Find the greatest displacement.

10 $v = 4t$. Find the acceleration.

Practice

MEDIUM · 10 questions

The same skill with more to simplify first.

1 $s = t^3 - 6t^2 + 9t$. Find v and a.

2 For that motion, find every instant at which the particle is at rest.

3 For that motion, find the acceleration at $t = 2$.

4 A balloon's radius grows at 0.5 cm/s. Find $\frac{dV}{dt}$ when $r = 10$ cm.

5 Water is poured into a cylinder of radius 5 cm at 50 cm³/s. Find the rate the level rises.

6 $P(x) = 50x - x^2 - 200$. Find the output giving greatest profit.

7 $C(x) = x^2 + 10x + 50$. Find the marginal cost at $x = 5$.

8 The side of a square grows at 2 cm/s. Find $\frac{dA}{dt}$ when the side is 6 cm.

9 $s = \sqrt{t + 1}$. Find v at $t = 3$.

10 $v = 6t - t^2$. Find the greatest velocity.

Challenge

HARD · 10 questions

Letters in place of numbers, and more than one idea at once.

1 $s = t^3 - 3t^2 - 9t + 5$, with $t \geq 0$. Find when the particle is at rest.

2 $s = t^2 - 4t$. Find the total distance travelled in the first 4 seconds.

3 A ladder 13 m long leans against a wall; its foot slides out at 2 m/s. Find the rate the top falls when the foot is 5 m out.

4 A conical tank has radius half its depth. Water drains at 8 cm³/s. Find the rate the depth falls when the depth is 4 cm.

5 A man 2 m tall walks at 1.5 m/s away from a lamp 6 m high. Find the rate his shadow lengthens.

6 A circular oil slick grows in area at 4 m²/s. Find $\frac{dr}{dt}$ when $r = 10$ m.

7 $R = 100x - 2x^2$ and $C = 20x + 100$. Find the output giving greatest profit.

8 $s = 2t^3 - 15t^2 + 24t$. Find the least velocity on $0 \le t \le 5$.

9 A snowball melts so that its volume falls at a rate proportional to its surface area. Show that the radius falls at a constant rate.

10 $v = t^2 - 5t + 6$. Find the distance travelled in the first 4 seconds.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1. $s = t^3 - 9t^2 + 24t$. Find the total distance travelled in the first 5 seconds.

2. Water flows at 10 cm³/s into a cone, apex down, whose radius equals its depth. Find the rate the depth rises when it is 5 cm.

3. A searchlight 100 m from a straight shore turns at 0.1 rad/s. Find the speed of the spot of light when the beam makes $30°$ with the perpendicular.

4. Show that profit is greatest where marginal revenue equals marginal cost.

5. Demand is $p = 100 - 2x$. Find the output giving greatest revenue, and the price there.

6 A kite flies 100 m above the ground and moves horizontally at 8 m/s. Find the rate the string pays out when 200 m is out.

7 Show that if $s = at^2 + bt + c$ the acceleration is constant, and identify it.

8 A particle has $a = 6t$, with $v(0) = 1$ and $s(0) = 2$. Find $s(t)$.

9 The side of an equilateral triangle grows at 2 cm/s. Find the rate its area grows when the side is 10 cm.

10 Two cars leave a crossroads at the same moment, one north at 60 km/h and one east at 80 km/h. Find the rate at which the distance between them grows.

11 The surface area of a cube grows at 12 cm²/s. Find the rate its volume grows when the edge is 5 cm.

12 For $s = t^n$ with $n \neq 0$, show that $\frac{as}{v^2}$ is the same constant at every instant.

13 A particle has $v = (1 + t)^2$ with $s(0) = 0$. Find $s(t)$, and the distance travelled in the first 2 seconds.

14 $C(x) = \frac{x^3}{3} - 10x^2 + 200x$. Find the output at which the marginal cost is least, and that least marginal cost.

15 A spherical balloon leaks at 2 cm³/s. Find the rate its surface area shrinks when the radius is 5 cm.

Revision · Control work 3 — the derivative in use

Tangents, turning points, optimisation, connected rates and approximation together — everything the derivative has been put to.

Lessons 35–36 · Control work 3 and work on the mistakes · 45 questions

KEY WORDS

- tangent
- stationary point
- optimisation
- connected rates
- approximation
- percentage error
- velocity
- acceleration
- justification

REMEMBER

1. Differentiate first, substitute second. Putting the number in early loses the function.
2. Connected rates: relate the quantities, differentiate with respect to t, substitute last.
3. Optimisation: objective, constraint, domain, $f'(x) = 0$, then justify the nature.
4. $\Delta y \approx f'(x)\Delta x$ holds for a short step only — the error grows like $(\Delta x)^2$.
5. Check the units and the sense of every answer before writing it down.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Differentiate $y = x^4$.

2. Differentiate $y = (2x + 1)^3$.

3. Find the tangent gradient to $y = x^2$ at $x = 5$.

4. $s = t^2 + t$. Find v at $t = 2$.

5. Estimate $(1.01)^2$.

6. Find the stationary point of $y = x^2 - 8x$.

7 Differentiate $y = \frac{2}{x^3}$.

8 Two numbers add to 20. Find their greatest product.

9 $s = 4t^3$. Find the acceleration.

10 Estimate $\sqrt{9.1}$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $y = x^3 - 12x$. Find the stationary points.

2 $y = x^3 - 12x$. Classify them.

3 Find the tangent to $y = x^3$ at $(2, 8)$.

4 A circle's radius grows at 3 cm/s. Find $\frac{dA}{dt}$ when $r = 4$ cm.

5 Estimate $(27.3)^{1/3}$.

6 A rectangle has perimeter 24 cm. Find its greatest area.

7 $y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ at $x = 4$.

8 $s = t^3 - 3t$, with $t \geq 0$. Find when the particle is at rest.

9 A radius is measured with a 0.5 % error. Find the percentage error in the volume of the sphere.

10 $y = x + \frac{4}{x}$. Find the stationary points.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $y = x^2 - 2x + 3$. Find the tangent parallel to $y = 4x$.

2 A closed cylinder has volume 54π cm³. Find the radius giving least surface area.

3 Estimate $\sin 29°$ to four decimal places.

4 $s = t^4 - 4t^3$. Find every instant at which the acceleration is zero.

5 A ladder 10 m long has its foot 6 m from the wall, sliding out at 1 m/s. Find the rate the top falls.

6 $y = x^2(x - 4)$. Find and classify the stationary points.

7 Find the greatest value of $y = x(8 - x^2)$ on $[0,\ 2\sqrt{2}]$.

8 Estimate $(4.02)^{3/2}$.

9 The volume of a cube grows at 12 cm³/s. Find the rate the edge grows when the edge is 2 cm.

10 Show that $y = \dfrac{x - 1}{x + 2}$ is increasing on each of its branches.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 $s = t^4 - 8t^2$. Find the total distance travelled in the first 3 seconds.

2 Find the greatest area of a rectangle with its base on the x-axis and its upper corners on $y = 9 - x^2$.

3 Show that the error in estimating $(1 + h)^n$ linearly is about $\frac{n(n-1)}{2}h^2$.

4 A trough 10 m long has a cross-section that is an isosceles triangle 1 m deep and 1 m across the top. Water flows in at 0.2 m³/min. Find the rate the depth rises when it is 0.5 m.

5 Prove that of all rectangles inscribed in a circle, the square has the greatest perimeter.

6 Find the shortest distance from $(0, 1)$ to the curve $y = x^2$.

7 A cone has fixed slant height L. Find its greatest volume.

8 Use the iteration $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ to solve $\cos x = x$ from $x_0 = 0.75$, to four decimal places.

9 Show that a cubic with a stationary point of inflection has exactly one real root.

10 A box with a square base and no lid has total surface area 108 cm². Find its greatest volume.

11 The period of a pendulum is $T = 2\pi\sqrt{\dfrac{L}{g}}$, with $g = 9.8$. Estimate the change in T when L goes from 1 m to 1.02 m.

12 $v = 3t^2 - 12t + 9$. Find the greatest speed on $0 \le t \le 4$.

13 Show that for a sphere $\frac{dV}{dr}$ equals the surface area, and say why that is to be expected.

14 Find a and b so that $y = ax^3 + bx$ has a stationary point at $(1, -2)$.

15 Two posts 6 m and 8 m high stand 10 m apart. A rope runs from the top of one to a point on the ground between them and up to the other. Find its least length.

The antiderivative and the indefinite integral

Differentiation run backwards — a whole family of curves, and the one condition that picks a single member out of it.

Lessons 37–40 · Algebra and Calculus 11, §2.1 · Pure Mathematics 1, 9.1 · 45 questions

KEY WORDS

- antiderivative
- indefinite integral
- constant of integration
- integrand
- family of curves
- initial condition
- particular solution
- integration
- inverse operation

REMEMBER

1. F is an antiderivative of f if $F' = f$. Checking one is always possible: differentiate it.
2. If F is an antiderivative then so is $F + C$, and every antiderivative has that form.
3. $\int f(x)dx = F(x) + C$. The $+ C$ is part of the answer, not decoration.
4. The family $F + C$ is a stack of parallel curves; an initial condition picks one out.
5. There is no product rule and no quotient rule for integration — simplify the integrand first.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\int x\,dx$.

2. Find $\int x^2\,dx$.

3. Find $\int 3\,dx$.

4. Find $\int x^5\,dx$.

5. Find $\int (2x + 1)\,dx$.

6. Find an antiderivative of $4x^3$.

7. Find $\int 0\ dx$.

8. Find $\int (x^2 - x)\ dx$.

9. $F'(x) = 2x$ and $F(0) = 5$. Find $F(x)$.

10. Differentiate $\frac{x^3}{3} + C$.

Practice

MEDIUM · 10 questions

The same skill with more to simplify first.

1. Find $\int (3x^2 - 4x + 1)\ dx$.

2. Find $\int \sqrt{x}\ dx$.

3. Find $\int x^{-2}\ dx$.

4. Find $\int (x + 1)^2\ dx$.

5. Find $\int \frac{1}{x^2}\ dx$.

6. $F'(x) = 3x^2$ and $F(1) = 4$. Find $F(x)$.

7 Find $\int(5 - 2x)\,dx$.

8 $v = 6t$ and $s(0) = 2$. Find $s(t)$.

9 Find $\int(x^3 + x)\,dx$.

10 A curve passes through $(1, 2)$ and has gradient $2x$. Find its equation.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\int(x^2 + 1)^2\,dx$.

2 Find $\int \frac{x^2 - 1}{x^2}\,dx$.

3 Find $\int(\sqrt{x} + \frac{1}{\sqrt{x}})\,dx$.

4 Find $\int x(x + 2)\,dx$.

5 $F'(x) = \dfrac{1}{x^2}$ and $F(1) = 0$. Find $F(x)$.

6 $a = 6t$, with $v(0) = 0$ and $s(0) = 0$. Find $s(t)$.

7 Find $\int (2x + 3)^4 \, dx$.

8 Find $\int \dfrac{1}{\sqrt{x}} \, dx$.

9 A curve has gradient $3x^2 - 6x$ and passes through $(2, 0)$. Find its equation.

10 Find $\int (x + 1)(x - 1) \, dx$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Prove that two antiderivatives of the same function differ by a constant.

2 Verify that $\int |x|\,dx = \frac{x|x|}{2} + C$.

3 Find f with $f'(x) = \frac{1}{x^2}$, $f(1) = 0$ and $f(-1) = 0$.

4 Show that $\int (1 + x)^n\,dx = \frac{(1 + x)^{n+1}}{n + 1} + C$ for $n \neq -1$, and say what happens at $n = -1$.

5 A curve has $\frac{d^2y}{dx^2} = 6x$, with $\frac{dy}{dx} = 1$ and $y = 2$ at $x = 0$. Find y.

6 Show that $\int f'(x) \cdot f(x)\,dx = \frac{[f(x)]^2}{2} + C$.

7 Find every curve whose gradient at each point is twice the x-coordinate, and the one through $(2, 1)$.

8 A ball is thrown upwards at 20 m/s from ground level, with $a = -10$ m/s². Find its greatest height.

9 Explain why $\int x^n \, dx = \dfrac{x^{n+1}}{n+1} + C$ fails for $n = -1$, and give the correct result.

10 Find $\int \dfrac{x^2 + 1}{x + 1} \, dx$.

11 Two curves have the same gradient function and meet at $(1, 3)$. Show that they are the same curve.

12 Find f with $f''(x) = 0$, $f(0) = 1$ and $f(1) = 4$.

13 A car starts from rest with $a = 12 - 6t$. Find when it is next at rest, and how far it has gone by then.

14 Describe the family $y = \frac{x^2}{2} + C$ geometrically, and say what every member shares.

15 F is an antiderivative of f and G of g. Show that $F + G$ is one of $f + g$, and that FG is generally not one of fg.

The table of integrals and the rules of integration

Every derivative read backwards, the linear substitution that extends them, and the rewriting that has to come first.

Lessons 41–46 · Algebra and Calculus 11, §2.2 · Pure Mathematics 1, 9.2–9.3 · 45 questions

KEY WORDS

- table of integrals
- power rule for integration
- linear substitution
- integrable form
- exceptional case
- term by term
- natural logarithm
- reciprocal function

REMEMBER

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$; and $\int \frac{dx}{x} = \ln|x| + C$, the exceptional case.
2. $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$ — the $\frac{1}{a}$ is the mark most often lost.
3. $\int \sin x\, dx = -\cos x + C$ and $\int \cos x\, dx = \sin x + C$: the minus sign sits on the opposite one from differentiation.
4. Constants come out and sums split term by term. There is no product or quotient rule.
5. Rewrite first — roots and reciprocals as powers, products expanded, fractions split.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\int x^4\, dx$.

2. Find $\int \frac{dx}{x}$.

3. Find $\int e^x\, dx$.

4. Find $\int \sin x\, dx$.

5. Find $\int \cos x\, dx$.

6 Find $\int (2x + 1)^3 \, dx$.

7 Find $\int e^{2x} \, dx$.

8 Find $\int \frac{dx}{x + 1}$.

9 Find $\int \sec^2 x \, dx$.

10 Find $\int 5x^2 \, dx$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Find $\int (3x + 2)^5 \, dx$.

2 Find $\int \sin 3x \, dx$.

3 Find $\int \cos(2x + 1) \, dx$.

4 Find $\int e^{-x} \, dx$.

5 Find $\int \frac{dx}{2x + 3}$.

6 Find $\int \sqrt{x + 4}\ dx$.

7 Find $\int (x^2 + \frac{2}{x})\ dx$.

8 Find $\int 3^x\ dx$.

9 Find $\int \frac{dx}{(3x - 1)^2}$.

10 Find $\int (\sin x - \cos x)\ dx$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\int (x + 1)(x - 2)\ dx$.

2 Find $\int \frac{x^2 + 1}{x}\ dx$.

3 Find $\int \sin^2 x\ dx$.

4 Find $\int \cos^2 x \, dx$.

5 Find $\int \tan^2 x \, dx$.

6 Find $\int (e^x + 1)^2 \, dx$.

7 Find $\int \frac{dx}{\sqrt{2x + 1}}$.

8 Find $\int \frac{2x}{x^2 + 1} \, dx$.

9 Find $\int x(x^2 + 1)^3 \, dx$.

10 Find $\int (1 + \tan^2 x) \, dx$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1. Prove the linear substitution rule $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$ for $n \neq -1$.

2. Find $\int \sin x \cdot \cos x \, dx$ in three different ways, and reconcile the answers.

3. Find $\int \frac{dx}{x^2 - 1}$.

4. Find $\int \frac{x}{x+1} \, dx$.

5. Find $\int \frac{x^3}{x^2+1} \, dx$.

6 Find $\int \sin^3 x \, dx$.

7 Find $\int \frac{e^x}{1 + e^x} \, dx$.

8 Show that $\int \frac{f'(x)}{f(x)} \, dx = \ln|f(x)| + C$.

9 Find $\int \frac{\ln x}{x} \, dx$.

10 Find $\int \frac{dx}{x \ln x}$.

11 Find $\int |x - 1|\, dx$ on each side of $x = 1$, and show that the two pieces can be made to join.

12 Show that $\int \cos^2 x\, dx + \int \sin^2 x\, dx = x + C$, and check it against the Challenge answers.

13 Find $\int \frac{2x + 3}{x^2 + 3x + 1}\, dx$.

14 A curve has $\frac{dy}{dx} = \sin x$ and passes through $(\frac{\pi}{2}, 0)$. Find its equation.

15 Find $\int (e^x - e^{-x})\, dx$, and say what the answer is a multiple of.

Revision · Control work 4 — the indefinite integral

The antiderivative, the table and the rules together — and the quarter behind them.

Lessons 47–48 · Control work 4 and the quarter review · 45 questions

KEY WORDS

- antiderivative
- indefinite integral
- constant of integration
- linear substitution
- initial condition
- term by term
- rewrite first
- check by differentiating

REMEMBER

1. Rewrite before integrating: expand, split, turn roots and reciprocals into powers.
2. Linear inner function: divide by a. Anything harder is Quarter III.
3. Every indefinite integral ends in $+ C$; every initial condition pins it down.
4. Reversing a second derivative needs two conditions, not one.
5. Differentiate your answer. It takes seconds and it settles the question.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\int x^3 \, dx$.

2. Find $\int dx$.

3. Find $\int e^{4x} \, dx$.

4. Find $\int \cos 4x \, dx$.

5. Find $\int (x + 2) \, dx$.

6. Find $\int \dfrac{dx}{x + 3}$.

7 Find $\int (3x - 1)^2 \, dx$.

8 Find $\int \sin(x + 1) \, dx$.

9 $F'(x) = 4x$ and $F(0) = 1$. Find $F(x)$.

10 Find $\int 2^x \, dx$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Find $\int (x^2 - 3x + 2) \, dx$.

2 Find $\int (4x + 1)^3 \, dx$.

3 Find $\int x\sqrt{x} \, dx$.

4 Find $\int e^{3x} \, dx$.

5 Find $\int \frac{dx}{x - 2}$.

6 Find $\int \sin 2x \, dx$.

7 $v = 4t + 1$ and $s(0) = 0$. Find $s(t)$.

8 Find $\int \frac{x - 1}{x}\,dx$.

9 Find $\int \sec^2 3x\,dx$.

10 A curve has gradient $\frac{1}{x^2}$ and passes through $(1, 2)$. Find its equation.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\int \frac{2x + 1}{x^2 + x + 1}\,dx$.

2 Find $\int (x + 1)^3\,dx$.

3 Find $\int (1 - x)^5\,dx$.

4 Find $\int (\sin x + \sec^2 x)\,dx$.

5 Find $\int x^2(x^3 + 1)^2\,dx$.

6 Find $\int \dfrac{dx}{1 - 2x}$.

7 Find $\int (e^x + e^{2x})\,dx$.

8 $a = 2$, with $v(0) = 3$ and $s(0) = 1$. Find $s(t)$.

9 Find $\int (\sqrt{x} - 1)^2\,dx$.

10 Find $\int \cos^2 \dfrac{x}{2}\,dx$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Find $\int x\sqrt{x + 1}\,dx$ by writing $u = x + 1$.

2 Find $\int \frac{dx}{e^x + 1}$.

3 Find $\int \sin^4 x \, dx$.

4 Verify by differentiation that $\int \sec x \, dx = \ln|\sec x + \tan x| + C$.

5 A ball is thrown upwards at 29.4 m/s from a height of 10 m, with $a = -9.8$ m/s². Find its greatest height.

6 Find $\int \frac{x^2 + 1}{(x + 1)^2} \, dx$.

7 Show that if f is odd then the antiderivative F with $F(0) = 0$ is even.

8 Find $\int \frac{dx}{x^2 + 4}$.

9 Verify by differentiation that $\int x\, e^x\, dx = e^x(x - 1) + C$.

10 A curve has $\frac{d^2y}{dx^2} = \sin x$, with $\frac{dy}{dx} = 0$ and $y = 0$ at $x = 0$. Find y.

11 Find $\int \frac{1 + x}{1 + x^2}\, dx$.

12 Show that $\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1} + C$ for $n \neq -1$, and use it on $\int x^2(x^3+5)^4\,dx$.

13 Water flows into an empty tank at $\frac{dV}{dt} = 6t - t^2$ litres per minute. Find the volume when the flow stops.

14 Find $\int \frac{dx}{\sqrt{4-x^2}}$.

15 Explain why $\int e^{x^2} dx$ has no answer in elementary functions, and say what is done instead.

Answers

Topic 1 · Approximate calculations with the derivative

Focus · easy

1. 0.06
2. 1.2
3. $\frac{1}{60} \approx 0.0167$
4. 10.05
5. 2.005
6. 4.04
7. 1.006
8. -0.025
9. 0.99
10. $\Delta y \approx f'(x) \cdot \Delta x$

Practice · medium

1. 5.1
2. 2.0083
3. 27.54
4. 0.4975
5. 2 cm²
6. 2 %
7. $0.5\pi \approx 1.57$ cm²
8. 2.16 cm³
9. 0.02
10. 0.01

Challenge · hard

1. 10.100
2. 10.0033
3. 0.515
4. 0.05
5. 1.02
6. 3 %
7. 4 %
8. 16.32
9. $1 + nx$
10. 0.99

Beyond · very hard

1 Of order $(\Delta x)^2$

Expand $f(a + h)$ as a Taylor series: $f(a) + f'(a)h + \frac{f''(a)}{2}h^2 + \ldots$. The linear estimate keeps the first two terms, so the error is $\frac{f''(a)}{2}h^2$ plus higher powers. Halving the step therefore cuts the error by a factor of about four — which is why the method is good for $\sqrt{101}$ and poor for $\sqrt{200}$.

2 7.0714; the true value is 7.07107, so the estimate is high by about 0.0004

Take $f(x) = \sqrt{x}$ at $x = 49$, where $f = 7$ and $f' = \frac{1}{14}$. Then $\sqrt{50} \approx 7 + \frac{1}{14} = 7.0714$. The error agrees with the previous question: $f'' = -\frac{1}{4x^{3/2}}$, so $\frac{|f''|}{2}h^2 \approx \frac{1}{8(343)} \approx 0.0004$.

3 $1 + \frac{x}{2}$; it gives 1.05 against the true 1.0488

With $f(x) = \sqrt{1 + x}$, $f(0) = 1$ and $f'(x) = \frac{1}{2\sqrt{1 + x}}$, so $f'(0) = \frac{1}{2}$. The tangent at the origin is $y = 1 + \frac{x}{2}$. At $x = 0.1$ it gives 1.05, against $\sqrt{1.1} = 1.0488$ — high by about 0.0012, as the curve is concave down.

4 The estimate is 2; the true value is 2.5937

With $f(x) = x^{10}$ at $x = 1$, $f' = 10$, so $(1.1)^{10} \approx 1 + 10(0.1) = 2$. The true value is 2.5937, so the estimate is out by 23 %. A step of 0.1 is not small when the tenth derivative is large: the method is only as good as the step is short.

5 0.5 %

T is proportional to $L^{1/2}$, and the relative error in L^n is n times the relative error in L. With $n = \frac{1}{2}$ the period lengthens by about 0.5 %. To slow a clock by 1 % the pendulum must be lengthened by 2 %.

6 6 %

$V = \frac{1}{3}\pi r^2 h$ with h fixed, so V is proportional to r^2 and the relative change doubles: about 6 %. Had the height varied instead, with r fixed, the change would have been 3 %, since V is only linear in h.

7 Always an underestimate

$f''(x) = \frac{2}{x^3} > 0$ for $x > 0$, so the curve is concave up and lies above every one of its tangents. The linear estimate reads off the tangent, so it falls short. The same argument makes the estimate of $\sqrt{x}$ an overestimate, since that curve is concave down.

8 1.4142

Here $f'(x) = 2x$, so the iteration is $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$. From $x_0 = 1.5$: $x_1 = 1.41667$, $x_2 = 1.414216$, $x_3 = 1.4142136$. The digits roughly double each step, which is what makes the method worth the extra derivative.

9 1.3247

$f'(x) = 3x^2 - 1$. From $x_0 = 1.5$: $f = 0.875$ and $f' = 5.75$, so $x_1 = 1.3478$. Then $f = 0.1007$ and $f' = 4.4501$, giving $x_2 = 1.3252$, and $x_3 = 1.32472$. This number is the plastic ratio, the real root of the equation.

10 The factor is n

With $y = x^n$, $\frac{dy}{dx} = nx^{n-1}$, so $\Delta y \approx nx^{n-1}\Delta x$. Dividing by $y = x^n$ gives $\frac{\Delta y}{y} \approx n\frac{\Delta x}{x}$. The result holds for any real n, so a 1 % error in a radius gives 2 % in an area, 3 % in a volume and 0.5 % in a square root.

11 0.5151

Work from $60° = \frac{\pi}{3}$, where $\cos = 0.5$ and the derivative is $-\sin 60° = -0.8660$. The step is $-1° = -0.01745$ radians, so $\cos 59° \approx 0.5 - 0.8660(-0.01745) = 0.5 + 0.0151 = 0.5151$. The true value is 0.51504. The step must be in radians even though the angle is quoted in degrees.

12 1.00995, against the linear 1.01

The linear estimate is $1 + 0.01 = 1.01$. The next term is $-\frac{(0.02)^2}{8} = -0.00005$, giving 1.00995. The true value is 1.0099505, so the quadratic term has cut the error from 5×10^{-5} to under 10^{-7}.

13 2 %

Here $\frac{1}{x^2} = x^{-2}$, so by the rule above the relative error is multiplied by $|-2| = 2$, giving 2 %. The sign of the index tells you the direction of the change but not its size: an overestimate of x gives an underestimate of $\frac{1}{x^2}$.

14 Exact for every linear f

If $f(x) = mx + c$ then $f'(x) = m$ for every x, and $f(a + h) = m(a + h) + c = (ma + c) + mh = f(a) + f'(a)h$. The estimate and the true value agree exactly, for any h. This matches the previous questions: the error term involves f'', and $f'' \equiv 0$ for a straight line.

15 $h \le 0.057$, roughly

The true value is $1 + 3h + 3h^2 + h^3$ and the linear estimate is $1 + 3h$, so the error is $3h^2 + h^3$. Setting $3h^2 + h^3 = 0.01$ and solving numerically gives $h \approx 0.0573$. At $h = 0.057$ the error is 0.00993; at $h = 0.058$ it is 0.01028.

Topic 2 · Modelling with the derivative

Focus · easy

1 $2t + 3$

2 7

3 $6t$

4 $t = 0$ and $t = 4$

5 10

6 metres per second

7 5

8 $20 - 2x$

9 25, at $t = 5$

10 4

Practice · medium

1 $v = 3t^2 - 12t + 9$, $a = 6t - 12$

2 $t = 1$ and $t = 3$

3 0

4 200π cm³/s

5 $\frac{2}{\pi}$ cm/s

6 $x = 25$

7 20

8 24 cm²/s

9 $\frac{1}{4}$

10 9, at $t = 3$

Challenge · hard

1 $t = 3$

2 8

3 $\frac{5}{6}$ m/s downwards

4 $\frac{2}{\pi}$ cm/s

5 0.75 m/s

6 $\frac{1}{5\pi}$ m/s

7 $x = 20$

8 -13.5, at $t = 2.5$

9 $\frac{dr}{dt}$ is constant

10 $\frac{17}{3}$

Beyond · very hard

1 28

$v = 3t^2 - 18t + 24 = 3(t - 2)(t - 4)$, so the particle rests at $t = 2$ and $t = 4$. The positions are $s(0) = 0$, $s(2) = 20$, $s(4) = 16$ and $s(5) = 20$. Adding the moduli of the changes: $20 + 4 + 4 = 28$. The displacement is only 20.

2 $\frac{2}{5\pi}$ cm/s

With $r = h$, $V = \frac{1}{3}\pi h^3$, so $\frac{dV}{dh} = \pi h^2$ and $\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}$. At $h = 5$: $10 = 25\pi \frac{dh}{dt}$, giving $\frac{dh}{dt} = \frac{2}{5\pi} \approx 0.127$ cm/s. The level slows as the cone widens, since the rate goes as $\frac{1}{h^2}$.

3 $\frac{40}{3} \approx 13.3$ m/s

With x measured along the shore from the foot of the perpendicular, $x = 100\tan\theta$. Then $\frac{dx}{dt} = 100\sec^2\theta \cdot \frac{d\theta}{dt}$. At $\theta = 30°$, $\sec^2\theta = \frac{4}{3}$, so $\frac{dx}{dt} = 100 \times \frac{4}{3} \times 0.1 = \frac{40}{3}$ m/s. The spot races away as θ approaches $90°$.

4 At $R'(x) = C'(x)$

Profit is $P(x) = R(x) - C(x)$, so $P'(x) = R'(x) - C'(x)$. A stationary point needs $P' = 0$, that is $R' = C'$. For it to be a maximum $P'' < 0$, so $R'' < C''$ — marginal revenue must be falling faster than marginal cost at that output.

5 $x = 25$ at a price of 50

Revenue is $R = px = 100x - 2x^2$, so $R' = 100 - 4x$, zero at $x = 25$, and $R'' = -4 < 0$, a maximum. The price is then $p = 100 - 50 = 50$ and the revenue 1250. Revenue is greatest at half the output that would drive the price to zero.

6 $4\sqrt{3} \approx 6.93$ m/s

With x the horizontal distance and L the string, $x^2 + 100^2 = L^2$. When $L = 200$, $x = \sqrt{30000} = 100\sqrt{3}$. Differentiating, $x\frac{dx}{dt} = L\frac{dL}{dt}$, so $\frac{dL}{dt} = \frac{100\sqrt{3} \times 8}{200} = 4\sqrt{3}$ m/s. It is less than 8, as it must be — only part of the motion is along the string.

7 $a_{acc} = 2a$

Differentiating twice gives $v = 2at + b$ and then $\frac{d^2s}{dt^2} = 2a$, free of t. So a quadratic displacement is exactly the motion of constant acceleration, with b the initial velocity and c the initial displacement. This is where the formula $s = ut + \frac{1}{2}at^2$ comes from.

8 $s = t^3 + t + 2$

Reversing the derivative, $v = 3t^2 + C$, and $v(0) = 1$ gives $C = 1$, so $v = 3t^2 + 1$. Reversing again, $s = t^3 + t + K$, and $s(0) = 2$ gives $K = 2$. Each reversal needs its own initial condition — the subject of Topic 4.

9 $10\sqrt{3} \approx 17.3$ cm²/s

The area is $A = \frac{\sqrt{3}}{4}s^2$, so $\frac{dA}{dt} = \frac{\sqrt{3}}{2}s\frac{ds}{dt}$. At $s = 10$ with $\frac{ds}{dt} = 2$ this is $\frac{\sqrt{3}}{2} \times 10 \times 2 = 10\sqrt{3}$ cm²/s.

10 100 km/h, constant

After t hours the distances are $60t$ and $80t$, so the separation is $d = \sqrt{(60t)^2 + (80t)^2} = 100t$. Hence $\frac{dd}{dt} = 100$ km/h at every instant. Because both motions are steady and perpendicular, the separation grows linearly — a 3–4–5 triangle scaled by $20t$.

11 15 cm³/s

With edge a, $S = 6a^2$ gives $12 = 12a\frac{da}{dt}$, so $\frac{da}{dt} = \frac{1}{a} = \frac{1}{5}$ cm/s. Then $V = a^3$ gives $\frac{dV}{dt} = 3a^2\frac{da}{dt} = 3 \times 25 \times \frac{1}{5} = 15$ cm³/s.

12 $\frac{n-1}{n}$

Here $v = nt^{n-1}$ and $a = n(n-1)t^{n-2}$. So $as = n(n-1)t^{2n-2}$ and $v^2 = n^2t^{2n-2}$. The powers of t cancel exactly, leaving $\frac{n-1}{n}$. For $n = 2$ — constant acceleration — it is $\frac{1}{2}$.

13 $s = \dfrac{(1+t)^3 - 1}{3}$; distance $\dfrac{26}{3}$

Reversing the derivative, $s = \dfrac{(1+t)^3}{3} + C$, and $s(0) = 0$ gives $C = -\dfrac{1}{3}$. At $t = 2$, $s = \dfrac{27-1}{3} = \dfrac{26}{3} \approx 8.67$. Since $v > 0$ throughout, distance and displacement agree.

14 $x = 10$, where the marginal cost is 100

The marginal cost is $C'(x) = x^2 - 20x + 200$. Differentiating again, $C''(x) = 2x - 20$, zero at $x = 10$, and $C''' = 2 > 0$, so it is a minimum of C'. The value is $100 - 200 + 200 = 100$. Note this is where $C'' = 0$ — the point of inflection of the cost curve.

15 0.8 cm²/s

$V = \dfrac{4}{3}\pi r^3$ gives $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$, so $-2 = 100\pi \dfrac{dr}{dt}$ and $\dfrac{dr}{dt} = -\dfrac{1}{50\pi}$. Then $S = 4\pi r^2$ gives $\dfrac{dS}{dt} = 8\pi r \dfrac{dr}{dt} = 8\pi \times 5 \times \left(-\dfrac{1}{50\pi}\right) = -0.8$ cm²/s. The π cancels, which is a useful check.

Topic 3 · Revision · Control work 3 — the derivative in use

Focus · easy

1 $4x^3$

2 $6(2x+1)^2$

3 10

4 5

5 1.02

6 $x = 4$

7 $-\dfrac{6}{x^4}$

8 100

9 $24t$

10 3.0167

Practice · medium

1 $x = 2$ and $x = -2$

2 Maximum at $x = -2$, minimum at $x = 2$

3 $y = 12x - 16$

4 24π cm²/s

5 3.0111

6 36 cm²

7 $\dfrac{4}{5}$

8 $t = 1$

9 1.5 %

10 $x = 2$ and $x = -2$

Challenge · hard

1 $y = 4x - 6$

2 $r = 3$ cm

3 0.4849

4 $t = 0$ and $t = 2$

5 0.75 m/s

6 Maximum at $x = 0$, minimum at $x = \dfrac{8}{3}$

7 $\dfrac{32\sqrt{6}}{9} \approx 8.71$

8 8.06

9 1 cm/s

10 $\dfrac{dy}{dx} = \dfrac{3}{(x+2)^2} > 0$

Beyond · very hard

1 41

$v = 4t^3 - 16t = 4t(t^2 - 4)$, so the particle rests at $t = 0$ and $t = 2$. The positions are $s(0) = 0$, $s(2) = 16 - 32 = -16$ and $s(3) = 81 - 72 = 9$. The distances are 16 and 25, totalling 41, while the displacement is only 9.

2 $12\sqrt{3} \approx 20.8$

With corners at $\pm x$ the area is $A = 2x(9 - x^2) = 18x - 2x^3$, for $0 < x < 3$. Then $A' = 18 - 6x^2 = 0$ gives $x = \sqrt{3}$, and $A'' = -12x < 0$, a maximum. The area is $2\sqrt{3}(9 - 3) = 12\sqrt{3}$.

3 $\dfrac{n(n-1)}{2}h^2$

The binomial expansion is $1 + nh + \dfrac{n(n-1)}{2}h^2 + \ldots$, and the linear estimate keeps only $1 + nh$. The leading term of what is left is $\dfrac{n(n-1)}{2}h^2$. For $n = 3$ and $h = 0.057$ this gives $3h^2 \approx 0.0097$, matching Topic 1.

4 0.04 m/min

At depth h the surface width is also h, by similar triangles, so the cross-section area is $\dfrac{h^2}{2}$ and $V = 5h^2$. Then $\dfrac{dV}{dt} = 10h\dfrac{dh}{dt}$, so $0.2 = 10(0.5)\dfrac{dh}{dt}$ and $\dfrac{dh}{dt} = 0.04$ m/min.

5 The square, of perimeter $2d\sqrt{2}$

With diameter d, a rectangle inscribed in the circle has sides $d\cos\theta$ and $d\sin\theta$. Its perimeter is $P = 2d(\cos\theta + \sin\theta) = 2d\sqrt{2}\sin(\theta + \dfrac{\pi}{4})$, which is greatest when $\theta = \dfrac{\pi}{4}$. Both sides are then $\dfrac{d}{\sqrt{2}}$, so the rectangle is a square.

6 $\dfrac{\sqrt{3}}{2} \approx 0.866$

Minimise $d^2 = x^2 + (x^2 - 1)^2$. Writing $u = x^2 \geq 0$ gives $d^2 = u + (u - 1)^2 = u^2 - u + 1$, least at $u = \dfrac{1}{2}$. Then $d^2 = \dfrac{1}{4} - \dfrac{1}{2} + 1 = \dfrac{3}{4}$, so $d = \dfrac{\sqrt{3}}{2}$. Two points are equally near, at $x = \pm\dfrac{1}{\sqrt{2}}$.

7 $\dfrac{2\pi L^3}{9\sqrt{3}}$, at height $\dfrac{L}{\sqrt{3}}$

With $r^2 + h^2 = L^2$, the volume is $V = \dfrac{1}{3}\pi(L^2 - h^2)h$. Then $\dfrac{dV}{dh} = \dfrac{\pi}{3}(L^2 - 3h^2) = 0$ gives $h = \dfrac{L}{\sqrt{3}}$ and $r^2 = \dfrac{2L^2}{3}$, so $V = \dfrac{\pi}{3} \cdot \dfrac{2L^2}{3} \cdot \dfrac{L}{\sqrt{3}} = \dfrac{2\pi L^3}{9\sqrt{3}}$.

8 0.7391

Take $f(x) = \cos x - x$, so $f'(x) = -\sin x - 1$. From $x_0 = 0.75$: $f = -0.01831$ and $f' = -1.68164$, giving $x_1 = 0.73911$. One more step gives $x_2 = 0.739085$. The value is the Dottie number, the only real solution.

9 Exactly one real root

A stationary inflection means $y' = 3a(x - p)^2$ for some p, since y' is a quadratic with a repeated root. Integrating, $y = a(x - p)^3 + k$. Setting $y = 0$ gives $(x - p)^3 = -\dfrac{k}{a}$, and a real number has exactly one real cube root, so there is exactly one real solution.

10 108 cm³

With base edge x and height h, $x^2 + 4xh = 108$ gives $h = \dfrac{108 - x^2}{4x}$. Then $V = x^2h = \dfrac{108x - x^3}{4}$, so $\dfrac{dV}{dx} = \dfrac{108 - 3x^2}{4} = 0$ at $x = 6$. Then $h = \dfrac{108 - 36}{24} = 3$ and $V = 36 \times 3 = 108$ cm³.

11 About 0.020 s longer

At $L = 1$, $T = 2\pi\sqrt{\dfrac{1}{9.8}} = 2.0071$ s. The relative change in T is half that in L, so $\dfrac{\Delta T}{T} \approx \dfrac{1}{2}(0.02) = 0.01$ and $\Delta T \approx 0.0201$ s. The clock loses about 1 % of its beats.

12 9

Speed is $|v|$. The velocity is a parabola with least value at $t = 2$, where $v = 12 - 24 + 9 = -3$. At the endpoints $v(0) = 9$ and $v(4) = 48 - 48 + 9 = 9$. The candidates for greatest speed are 9, 9 and $|-3| = 3$, so the greatest speed is 9.

13 $\dfrac{dV}{dr} = 4\pi r^2 = S$

$V = \dfrac{4}{3}\pi r^3$, so $\dfrac{dV}{dr} = 4\pi r^2$, which is the surface area. Geometrically, growing the radius by δr adds a thin shell of volume about $S \cdot \delta r$, so the rate of change of volume per unit radius is the area of the surface being coated. The same holds for a circle: $\dfrac{dA}{dr} = 2\pi r$, the circumference.

14 $a = 1$, $b = -3$

$\dfrac{dy}{dx} = 3ax^2 + b$, and a stationary point at $x = 1$ needs $3a + b = 0$, so $b = -3a$. Passing through $(1, -2)$ gives $a + b = -2$, that is $a - 3a = -2a = -2$, so $a = 1$ and $b = -3$. The curve is $y = x^3 - 3x$.

15 $2\sqrt{74} \approx 17.2$ m

With the point x from the shorter post, $L = \sqrt{36 + x^2} + \sqrt{64 + (10 - x)^2}$. Setting $\dfrac{dL}{dx} = 0$ gives $\dfrac{x}{\sqrt{36 + x^2}} = \dfrac{10 - x}{\sqrt{64 + (10 - x)^2}}$ — the two ropes make equal angles with the ground — so $\dfrac{x}{6} = \dfrac{10 - x}{8}$ and $x = \dfrac{30}{7}$. Reflecting one post below ground makes it a straight line of length $\sqrt{(6 + 8)^2 + 10^2} = \sqrt{296} = 2\sqrt{74}$.

Topic 4 · The antiderivative and the indefinite integral

Focus · easy

1 $\dfrac{x^2}{2} + C$

2 $\dfrac{x^3}{3} + C$

3 $3x + C$

4 $\dfrac{x^6}{6} + C$

5 $x^2 + x + C$

6 x^4

7 C

8 $\dfrac{x^3}{3} - \dfrac{x^2}{2} + C$

9 $x^2 + 5$

10 x^2

Practice · medium

1 $x^3 - 2x^2 + x + C$

2 $\dfrac{2}{3}x^{3/2} + C$

6 $x^3 + 3$

7 $5x - x^2 + C$

8 $3t^2 + 2$

3 $-\dfrac{1}{x} + C$

4 $\dfrac{(x+1)^3}{3} + C$

5 $-\dfrac{1}{x} + C$

9 $\dfrac{x^4}{4} + \dfrac{x^2}{2} + C$

10 $y = x^2 + 1$

Challenge · hard

1 $\dfrac{x^5}{5} + \dfrac{2x^3}{3} + x + C$

2 $x + \dfrac{1}{x} + C$

3 $\dfrac{2}{3}x^{3/2} + 2\sqrt{x} + C$

4 $\dfrac{x^3}{3} + x^2 + C$

5 $1 - \dfrac{1}{x}$

6 t^3

7 $\dfrac{(2x+3)^5}{10} + C$

8 $2\sqrt{x} + C$

9 $y = x^3 - 3x^2 + 4$

10 $\dfrac{x^3}{3} - x + C$

Beyond · very hard

1 They differ by a constant

Let $F' = G' = f$ on an interval and put $H = F - G$. Then $H' = 0$ throughout. By the mean value theorem, for any two points $a < b$ there is c between them with $H(b) - H(a) = H'(c)(b - a) = 0$, so H takes the same value everywhere. Hence $F = G + C$. The interval matters — see question 3.

2 Correct for every x

Differentiate the proposed answer. From Quarter I, $\dfrac{d}{dx}(x|x|) = 2|x|$, so $\dfrac{d}{dx}\dfrac{x|x|}{2} = |x|$, which is the integrand. Writing it as two cases also works: $\dfrac{x^2}{2}$ for $x \geq 0$ and $-\dfrac{x^2}{2}$ for $x < 0$, which join smoothly at the origin.

3 $f(x) = 1 - \dfrac{1}{x}$ for $x > 0$ and $f(x) = -1 - \dfrac{1}{x}$ for $x < 0$

The antiderivative is $-\dfrac{1}{x} + C$, but the domain is split at 0, so each branch carries its own constant. On $x > 0$: $-1 + C_1 = 0$ gives $C_1 = 1$. On $x < 0$: $1 + C_2 = 0$ gives $C_2 = -1$. "Antiderivatives differ by a constant" holds on an interval, and this domain is not one.

4 At $n = -1$ the answer is $\ln|1 + x| + C$

Differentiating $\dfrac{(1+x)^{n+1}}{n+1}$ gives $(1+x)^n$ by the chain rule, since the inner derivative is 1. At $n = -1$ the formula divides by zero and fails; the integrand is then $\dfrac{1}{1+x}$, whose antiderivative is $\ln|1 + x| + C$.

5 $y = x^3 + x + 2$

Integrating once, $\dfrac{dy}{dx} = 3x^2 + C_1$, and the condition at $x = 0$ gives $C_1 = 1$. Integrating again, $y = x^3 + x + C_2$, and $y(0) = 2$ gives $C_2 = 2$. Each integration needs its own condition — two in all for a second derivative.

6 $\dfrac{[f(x)]^2}{2} + C$

Differentiate the right-hand side by the chain rule: $\dfrac{d}{dx}\dfrac{[f(x)]^2}{2} = f(x) \cdot f'(x)$, which is the integrand. The pattern generalises to $\int f' \cdot f^n \, dx = \dfrac{f^{n+1}}{n+1} + C$ — recognising it saves a substitution.

7 $y = x^2 + C$; through $(2, 1)$ it is $y = x^2 - 3$

The condition is $\dfrac{dy}{dx} = 2x$, so $y = x^2 + C$ — a stack of parallel parabolas. Through $(2, 1)$: $1 = 4 + C$ gives $C = -3$. Exactly one member passes through any given point, which is what the constant is for.

8 20 m, at $t = 2$ s

Integrating, $v = -10t + 20$ using $v(0) = 20$, and $s = -5t^2 + 20t$ using $s(0) = 0$. The greatest height is where $v = 0$, at $t = 2$, giving $s = -20 + 40 = 20$ m. It returns to the ground at $t = 4$, by the symmetry of the parabola.

9 It divides by zero; the answer is $\ln|x| + C$

At $n = -1$ the formula would give $\dfrac{x^0}{0}$, which is undefined. The integrand is $\dfrac{1}{x}$, and the function whose derivative is $\dfrac{1}{x}$ is $\ln|x|$, so $\int \dfrac{dx}{x} = \ln|x| + C$. The modulus matters: it makes the result valid for negative x as well.

10 $\dfrac{x^2}{2} - x + 2\ln|x + 1| + C$

The fraction is improper, so divide first: $x^2 + 1 = (x + 1)(x - 1) + 2$, giving $\dfrac{x^2 + 1}{x + 1} = x - 1 + \dfrac{2}{x + 1}$.

Integrating term by term gives $\dfrac{x^2}{2} - x + 2\ln|x + 1| + C$. Dividing before integrating is the standard first move on any improper algebraic fraction.

11 They coincide

If $y_1' = y_2'$ on an interval then $y_1 - y_2$ has zero derivative, so it is a constant k. At $x = 1$ both equal 3, so $k = 0$ and $y_1 \equiv y_2$. Two members of the same family either coincide or never meet — which is why the curves of $y = x^2 + C$ are parallel.

12 $f(x) = 3x + 1$

$f'' = 0$ gives $f' = C_1$, a constant, and then $f = C_1x + C_2$ — a straight line, as expected when the gradient never changes. From $f(0) = 1$, $C_2 = 1$; from $f(1) = 4$, $C_1 + 1 = 4$ so $C_1 = 3$. Here the two conditions are values rather than a value and a gradient, but two are still needed.

13 At $t = 4$ s, having travelled 32 m

Integrating with $v(0) = 0$ gives $v = 12t - 3t^2 = 3t(4 - t)$, which is zero again at $t = 4$. Integrating again with $s(0) = 0$ gives $s = 6t^2 - t^3$, so $s(4) = 96 - 64 = 32$ m. The velocity stays positive on $(0, 4)$, so distance and displacement agree.

14 Parallel parabolas, all with the same gradient at each x

Each member is the same parabola shifted vertically by C. At any fixed x every member has gradient x, so their tangents there are parallel. Exactly one member passes through each point of the plane, and no two ever meet — the constant counts the vertical shift and nothing else.

15 Sums carry over; products do not

Differentiating, $(F + G)' = F' + G' = f + g$, so the sum rule carries straight across. But $(FG)' = F'G + FG' = fG + Fg$, which is not fg in general. Take $f = g = 1$, so $F = G = x$: then $FG = x^2$ has derivative $2x$, not 1. This is exactly why integration has no product rule.

Topic 5 · The table of integrals and the rules of integration

Focus · easy

1. $\frac{x^5}{5} + C$
2. $\ln|x| + C$
3. $e^x + C$
4. $-\cos x + C$
5. $\sin x + C$
6. $\frac{(2x+1)^4}{8} + C$
7. $\frac{e^{2x}}{2} + C$
8. $\ln|x+1| + C$
9. $\tan x + C$
10. $\frac{5x^3}{3} + C$

Practice · medium

1. $\frac{(3x+2)^6}{18} + C$
2. $-\frac{\cos 3x}{3} + C$
3. $\frac{\sin(2x+1)}{2} + C$
4. $-e^{-x} + C$
5. $\frac{1}{2}\ln|2x+3| + C$
6. $\frac{2}{3}(x+4)^{3/2} + C$
7. $\frac{x^3}{3} + 2\ln|x| + C$
8. $\frac{3^x}{\ln 3} + C$
9. $-\frac{1}{3(3x-1)} + C$
10. $-\cos x - \sin x + C$

Challenge · hard

1. $\frac{x^3}{3} - \frac{x^2}{2} - 2x + C$
2. $\frac{x^2}{2} + \ln|x| + C$
3. $\frac{x}{2} - \frac{\sin 2x}{4} + C$
4. $\frac{x}{2} + \frac{\sin 2x}{4} + C$
5. $\tan x - x + C$
6. $\frac{e^{2x}}{2} + 2e^x + x + C$
7. $\sqrt{2x+1} + C$
8. $\ln(x^2+1) + C$
9. $\frac{(x^2+1)^4}{8} + C$
10. $\tan x + C$

Beyond · very hard

1 Proved by differentiating

Differentiate the proposed answer by the chain rule: $\frac{d}{dx}\frac{(ax+b)^{n+1}}{a(n+1)} = \frac{(n+1)(ax+b)^n \cdot a}{a(n+1)} = (ax+b)^n$, the integrand. The a in the denominator is exactly what cancels the a the chain rule brings down.

2 $\frac{\sin^2 x}{2} + C$, $-\frac{\cos^2 x}{2} + C$, $-\frac{\cos 2x}{4} + C$ — all the same family

Taking $f = \sin x$ gives $\frac{\sin^2 x}{2}$; taking $f = \cos x$ gives $-\frac{\cos^2 x}{2}$; using $\sin x \cos x = \frac{\sin 2x}{2}$ gives $-\frac{\cos 2x}{4}$.

They differ only by constants: $\frac{\sin^2 x}{2} = -\frac{\cos^2 x}{2} + \frac{1}{2}$, and $-\frac{\cos 2x}{4} = \frac{\sin^2 x}{2} - \frac{1}{4}$. All three are correct — the arbitrary constant absorbs the difference.

3 $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

Split into partial fractions: $\frac{1}{x^2-1} = \frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right)$. Integrating gives $\frac{1}{2}(\ln|x-1| - \ln|x+1|)$, which combines into $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$.

4 $x - \ln|x+1| + C$

The fraction is improper in the sense that the degrees match, so divide: $\frac{x}{x+1} = 1 - \frac{1}{x+1}$. Integrating term by term gives $x - \ln|x+1| + C$. Never attempt such a fraction without dividing first.

5 $\frac{x^2}{2} - \frac{1}{2}\ln(x^2+1) + C$

Divide: $\frac{x^3}{x^2+1} = x - \frac{x}{x^2+1}$. The first term integrates to $\frac{x^2}{2}$. For the second, the numerator is half the derivative of the denominator, so it gives $\frac{1}{2}\ln(x^2+1)$. No modulus is needed, since $x^2 + 1 > 0$.

6 $-\cos x + \frac{\cos^3 x}{3} + C$

Peel off one factor: $\sin^3 x = \sin x(1 - \cos^2 x)$. With $u = \cos x$ and $du = -\sin x\,dx$ the integral becomes $-\int(1 - u^2)du = -u + \frac{u^3}{3}$, that is $-\cos x + \frac{\cos^3 x}{3} + C$. An odd power always yields to this trick.

7 $\ln(1 + e^x) + C$

The numerator is exactly the derivative of the denominator, so the integral is $\ln|1 + e^x| + C$, and the modulus may be dropped since $1 + e^x > 0$ always.

8 $\ln|f(x)| + C$

Differentiate the right-hand side: by the chain rule $\frac{d}{dx}\ln|f(x)| = \frac{f'(x)}{f(x)}$, which is the integrand. Recognising this pattern on sight is worth more than any substitution — it handles $\tan x$, $\frac{2x}{x^2+1}$ and $\frac{e^x}{1+e^x}$ in one step each.

9 $\frac{(\ln x)^2}{2} + C$

Here $\frac{1}{x}$ is the derivative of $\ln x$, so with $f = \ln x$ the integrand is $f \cdot f'$, whose integral is $\frac{f^2}{2}$ by Topic 4. Hence the answer is $\frac{(\ln x)^2}{2} + C$.

10 $\ln|\ln x| + C$

Write the integrand as $\frac{1/x}{\ln x}$, which is $\frac{f'}{f}$ with $f = \ln x$. By the rule above the integral is $\ln|\ln x| + C$. Compare with the previous question: the same two pieces, one as a product and one as a quotient, give quite different answers.

11 $\frac{(x-1)^2}{2} + C$ for $x \geq 1$ and $-\frac{(x-1)^2}{2} + C$ for $x < 1$

For $x \geq 1$ the integrand is $x - 1$, giving $\frac{(x-1)^2}{2}$; for $x < 1$ it is $1 - x$, giving $-\frac{(x-1)^2}{2}$. Both vanish at $x = 1$, so taking the same constant on each side makes the antiderivative continuous — and differentiable, since both one-sided derivatives there are 0.

12 They sum to $x + C$

Since $\sin^2 x + \cos^2 x = 1$, the two integrands add to 1, whose integral is $x + C$. The Challenge answers were $\frac{x}{2} + \frac{\sin 2x}{4}$ and $\frac{x}{2} - \frac{\sin 2x}{4}$; adding them, the $\sin 2x$ terms cancel and $\frac{x}{2} + \frac{x}{2} = x$. A quick and complete check on both.

13 $\ln|x^2 + 3x + 1| + C$

The numerator is exactly the derivative of the denominator, so the $\frac{f'}{f}$ rule applies directly and the integral is $\ln|x^2 + 3x + 1| + C$. The modulus is needed here: the quadratic has real roots, so it does take negative values.

14 $y = -\cos x$

Integrating gives $y = -\cos x + C$. At $x = \frac{\pi}{2}$, $\cos x = 0$, so $0 = 0 + C$ and $C = 0$. The curve is $y = -\cos x$, which is the sine curve shifted a quarter period.

15 $e^x + e^{-x} + C$, which is $2\cosh x + C$

Integrating term by term: $\int e^x dx = e^x$ and $\int e^{-x} dx = -e^{-x}$, so the difference integrates to $e^x + e^{-x}$. Since $\cosh x = \frac{e^x + e^{-x}}{2}$, the answer is $2\cosh x + C$ — the integrand being $2\sinh x$, which mirrors $\int \sin$ and $\int \cos$ without the sign changes.

Topic 6 · Revision · Control work 4 — the indefinite integral

Focus · easy

1 $\frac{x^4}{4} + C$

2 $x + C$

3 $\frac{e^{4x}}{4} + C$

4 $\frac{\sin 4x}{4} + C$

5 $\frac{x^2}{2} + 2x + C$

6 $\ln|x + 3| + C$

7 $\frac{(3x - 1)^3}{9} + C$

8 $-\cos(x + 1) + C$

9 $2x^2 + 1$

10 $\frac{2^x}{\ln 2} + C$

Practice · medium

1 $\frac{x^3}{3} - \frac{3x^2}{2} + 2x + C$

2 $\frac{(4x + 1)^4}{16} + C$

3 $\frac{2}{5}x^{5/2} + C$

4 $\frac{e^{3x}}{3} + C$

5 $\ln|x - 2| + C$

6 $-\frac{\cos 2x}{2} + C$

7 $2t^2 + t$

8 $x - \ln|x| + C$

9 $\frac{\tan 3x}{3} + C$

10 $y = 3 - \frac{1}{x}$

Challenge · hard

1 $\ln|x^2 + x + 1| + C$

2 $\dfrac{(x+1)^4}{4} + C$

3 $-\dfrac{(1-x)^6}{6} + C$

4 $-\cos x + \tan x + C$

5 $\dfrac{(x^3+1)^3}{9} + C$

6 $-\dfrac{1}{2}\ln|1 - 2x| + C$

7 $e^x + \dfrac{e^{2x}}{2} + C$

8 $t^2 + 3t + 1$

9 $\dfrac{x^2}{2} - \dfrac{4}{3}x^{3/2} + x + C$

10 $\dfrac{x}{2} + \dfrac{\sin x}{2} + C$

Beyond · very hard

1 $\dfrac{2}{5}(x+1)^{5/2} - \dfrac{2}{3}(x+1)^{3/2} + C$

With $u = x + 1$ we have $x = u - 1$ and $dx = du$, so the integral becomes $\int (u-1)\sqrt{u}\,du = \int (u^{3/2} - u^{1/2})du = \dfrac{2}{5}u^{5/2} - \dfrac{2}{3}u^{3/2} + C$. Substituting back gives the answer. Replacing *everything* in terms of u is what makes the substitution work.

2 $x - \ln(1 + e^x) + C$

Write $\dfrac{1}{e^x + 1} = 1 - \dfrac{e^x}{e^x + 1}$, which is easy to check by combining the right-hand side. The first term integrates to x and the second, being $\dfrac{f'}{f}$, to $\ln(1 + e^x)$. Hence the result.

3 $\dfrac{3x}{8} - \dfrac{\sin 2x}{4} + \dfrac{\sin 4x}{32} + C$

Use $\sin^2 x = \dfrac{1 - \cos 2x}{2}$ twice. First $\sin^4 x = \dfrac{1}{4}(1 - 2\cos 2x + \cos^2 2x)$, then $\cos^2 2x = \dfrac{1 + \cos 4x}{2}$, giving $\dfrac{1}{4}\left(\dfrac{3}{2} - 2\cos 2x + \dfrac{\cos 4x}{2}\right)$. Integrating term by term gives the stated answer. Even powers always need the double-angle route.

4 Verified

Differentiating, $\dfrac{d}{dx}\ln|\sec x + \tan x| = \dfrac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} = \dfrac{\sec x(\tan x + \sec x)}{\sec x + \tan x} = \sec x$. The bracket cancels exactly, which is the small miracle that makes this the standard answer.

5 54.1 m, at $t = 3$ s

Integrating, $v = -9.8t + 29.4$ and $s = -4.9t^2 + 29.4t + 10$. The greatest height is where $v = 0$, at $t = 3$, giving $s = -44.1 + 88.2 + 10 = 54.1$ m.

6 $x - 2\ln|x+1| - \dfrac{2}{x+1} + C$

Put $u = x + 1$, so $x = u - 1$ and $x^2 + 1 = u^2 - 2u + 2$. The integrand becomes $1 - \dfrac{2}{u} + \dfrac{2}{u^2}$, which integrates to $u - 2\ln|u| - \dfrac{2}{u}$. Substituting back and absorbing the constant 1 gives the answer.

7 F is even

Consider $G(x) = F(-x)$. Then $G'(x) = -F'(-x) = -f(-x) = f(x)$, since f is odd. So G and F have the same derivative and differ by a constant; at $x = 0$ both are 0, so the constant is zero and $F(-x) = F(x)$. For example $f = x$ is odd and $F = \frac{x^2}{2}$ is even.

8 $\frac{1}{2}\arctan\frac{x}{2} + C$

Use the standard form $\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\arctan\frac{x}{a} + C$ with $a = 2$. Checking: $\frac{d}{dx}\frac{1}{2}\arctan\frac{x}{2} = \frac{1}{2} \cdot \frac{1/2}{1 + x^2/4} = \frac{1}{4 + x^2}$, as required.

9 Verified

Differentiate by the product rule: $\frac{d}{dx}[e^x(x - 1)] = e^x(x - 1) + e^x \cdot 1 = e^x(x - 1 + 1) = x\,e^x$, the integrand. Finding this answer rather than checking it needs integration by parts, which is Quarter III.

10 $y = x - \sin x$

Integrating once, $\frac{dy}{dx} = -\cos x + C_1$, and the condition at $x = 0$ gives $-1 + C_1 = 0$, so $C_1 = 1$. Integrating again, $y = -\sin x + x + C_2$, and $y(0) = 0$ gives $C_2 = 0$. The curve $y = x - \sin x$ is the cycloid's x-component, and it is increasing everywhere.

11 $\arctan x + \frac{1}{2}\ln(1 + x^2) + C$

Split the fraction: $\frac{1}{1 + x^2} + \frac{x}{1 + x^2}$. The first is the standard arctangent form. In the second the numerator is half the derivative of the denominator, giving $\frac{1}{2}\ln(1 + x^2)$. Splitting a sum in the numerator is always allowed; splitting one in the denominator never is.

12 $\frac{(x^3 + 5)^5}{15} + C$

Differentiating $\frac{f^{n+1}}{n + 1}$ by the chain rule gives $f^n f'$, the integrand. For the example, $f = x^3 + 5$ has $f' = 3x^2$, so $x^2 = \frac{f'}{3}$ and the integral is $\frac{1}{3} \cdot \frac{(x^3 + 5)^5}{5} = \frac{(x^3 + 5)^5}{15} + C$.

13 36 litres, at $t = 6$ min

The flow stops when $6t - t^2 = t(6 - t) = 0$, that is at $t = 6$ (the root $t = 0$ is the start). Integrating with $V(0) = 0$ gives $V = 3t^2 - \frac{t^3}{3}$, so $V(6) = 108 - 72 = 36$ litres. After $t = 6$ the rate turns negative and the tank would drain.

14 $\arcsin\frac{x}{2} + C$

Use the standard form $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\frac{x}{a} + C$ with $a = 2$. Checking: $\frac{d}{dx}\arcsin\frac{x}{2} = \frac{1/2}{\sqrt{1 - x^2/4}} = \frac{1}{\sqrt{4 - x^2}}$. The integral is defined only for $|x| < 2$.

15 No elementary antiderivative exists; it is computed numerically or named

Differentiation is an algorithm: every elementary function has an elementary derivative. Integration is a search, and the elementary functions are not closed under it — Liouville proved that no combination of powers, roots, exponentials, logarithms and trigonometric functions differentiates to e^{x^2}. In practice the integral is evaluated numerically, or given a name: the closely related $\int e^{-x^2} dx$ is the error function, on which the normal distribution rests.